Q1. EU Crypto-Asset Law (MiCA – April 2023)

- EU passed MiCA (Markets in Crypto-Assets) regulation.
- It controls crypto trading, holding, and service providers.
- Crypto companies must register and follow rules.
- Focus on consumer protection and preventing fraud.
- Reduces risks like money laundering.

Q2. Waiting for Transaction Confirmations

- A confirmation means the transaction is added to a block.
- More confirmations mean more security.
- It prevents double-spending attacks.
- Bitcoin usually waits for 6 confirmations.
- Helps ensure the transaction cannot be reversed.

Q3. ICO (Initial Coin Offering)

- ICO is a way to raise money for crypto projects.
- Investors buy tokens before the project is complete.
- Positive: Easy fundraising for startups.
- Negative: High risk of scams.
- Not always regulated.

Q4. Are Bitcoin & Ethereum Anonymous?

- This statement is false.
- They are pseudonymous, not anonymous.
- Wallet addresses are visible on the blockchain.
- Transactions can be traced.
- Real identity may be linked later.

Q5. Game Theory in Blockchain

- Prisoner’s Dilemma: Miners act honestly for rewards.
- Nash Equilibrium: Honest mining is the best strategy.
- Incentives encourage correct behavior.
- Punishments discourage cheating.
- Helps keep the network secure.

Q6. Alt-Coins

- Alt-coins are cryptocurrencies other than Bitcoin.
- Ethereum: Smart contracts and dApps.
- Ripple (XRP): Fast cross-border payments.
- They solve different problems.

- Offer innovation beyond Bitcoin.

Q7. Ethereum Technology for Property Rights

- NFTs (Non-Fungible Tokens).
- They show digital ownership.
- Used for art, land, music.
- Ownership is verifiable.
- Cannot be easily copied.

Q8. Hyperledger Fabric

- Permissioned blockchain (private).
- Used by businesses.
- No mining like Bitcoin.
- Faster and more scalable.
- Example: IBM Food Trust.

Q9. Web3 & CBDC Positives and Negatives

- Positive: Faster payments.
- Positive: Financial inclusion.
- Negative: Privacy concerns.
- Negative: Government control.
- Technology still evolving.

Q10. Scalability Issues in BTC & ETH

- Bitcoin has slow transactions.
- Ethereum has high gas fees.
- Causes network congestion.
- Lightning Network helps Bitcoin.
- Layer 2 solutions help Ethereum.

Q11. Technologies Behind Bitcoin (2009)

- Cryptography ensures security.
- Peer-to-peer networks remove middlemen.
- Hash functions link blocks securely.
- Merkle Trees store transactions efficiently.
- Satoshi Nakamoto combined these ideas.

Q12. Proof of Work (PoW)

- Miners solve complex math problems.

- First miner adds the block.
- Requires high energy.
- More secure but less eco-friendly.
- PoS uses validators instead of miners.